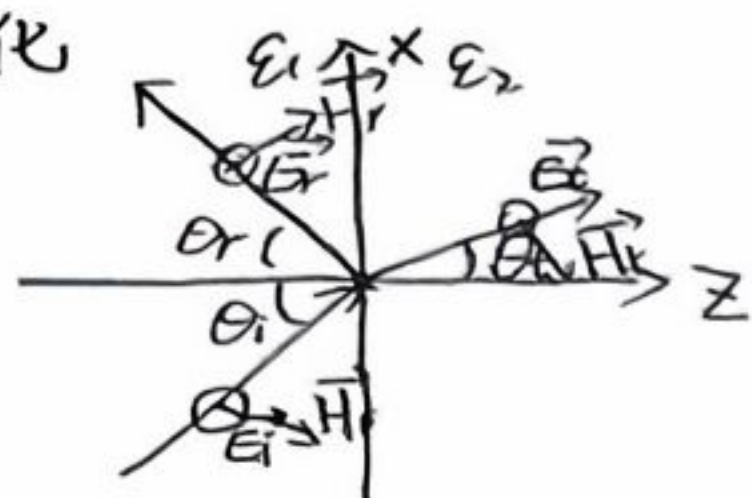


1. 射入射 (系数的推导)

垂直极化



-1

由边界条件 $E_{it} = E_{et}$

$$\vec{E}_{im} + \vec{E}_{rm} = \vec{E}_{tm}$$

$$H_{it} = H_{et} \quad (H_{im} - H_{rm}) \cos \theta_i = H_{tm} \cos \theta_t$$

$$m = \frac{1}{\eta_1} E_{im} \quad H_{rm} = \frac{1}{\eta_1} E_{rm}$$

$$H_{tm} = \frac{1}{\eta_2} E_{tm}$$

$$(-E_{im} - \frac{1}{\eta_1} E_{rm}) \cos \theta_i = \frac{1}{\eta_2} E_{tm} \cos \theta_t$$

$$\text{解得 } \Gamma_{\perp} = \frac{E_r}{E_i} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

$$T_{\perp} = \frac{E_t}{E_i} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

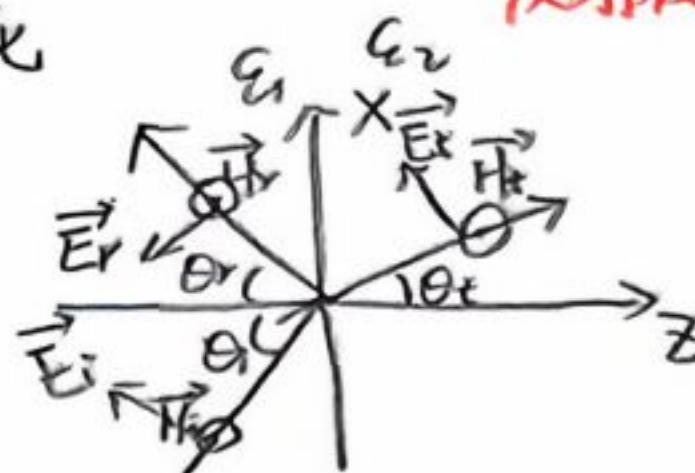
$$\text{若 } \mu_1 = \mu_2 = \mu \quad \Gamma_{\perp} = \frac{\sqrt{\epsilon_1} \cos \theta_i - \sqrt{\epsilon_2} \cos \theta_t}{\sqrt{\epsilon_1} \cos \theta_i + \sqrt{\epsilon_2} \cos \theta_t}$$

$$= \frac{\sin \theta_t \cos \theta_i - \sin \theta_i \cos \theta_t}{\sin \theta_t \cos \theta_i + \sin \theta_i \cos \theta_t} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$$

$$T_{\perp} = \frac{2 \cos \theta_i \sin \theta_t}{\sin(\theta_t + \theta_i)}$$

平行极化

反射波振幅比值



边界条件 $E_{it} = E_{et} \quad (E_{im} - E_{rm}) \cos \theta_i = E_{tm} \cos \theta_t$

$$H_{it} = H_{et} \quad H_{im} + H_{rm} = H_{tm}$$

$$H_{im} = \frac{1}{\eta_1} E_{im} \quad H_{rm} = \frac{1}{\eta_1} E_{rm}$$

$$H_{tm} = \frac{1}{\eta_2} E_{tm}$$

$$\eta_1 H_{im} - \eta_1 H_{rm} \cos \theta_i = \eta_2 H_{tm} \cos \theta_t$$

$$H_{im} (\eta_1 \cos \theta_i - \eta_2 \cos \theta_t) = H_{rm} (\eta_2 \cos \theta_t + \eta_1 \cos \theta_i)$$

$$\frac{H_{rm}}{H_{im}} = \frac{\eta_1 \cos \theta_i - \eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

$$\Gamma_{\parallel} = \frac{E_{rm}}{E_{im}} = \frac{H_{rm}}{H_{im}} = \frac{\eta_1 \cos \theta_i - \eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

$$T_{\parallel} = 1 + \Gamma_{\parallel} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

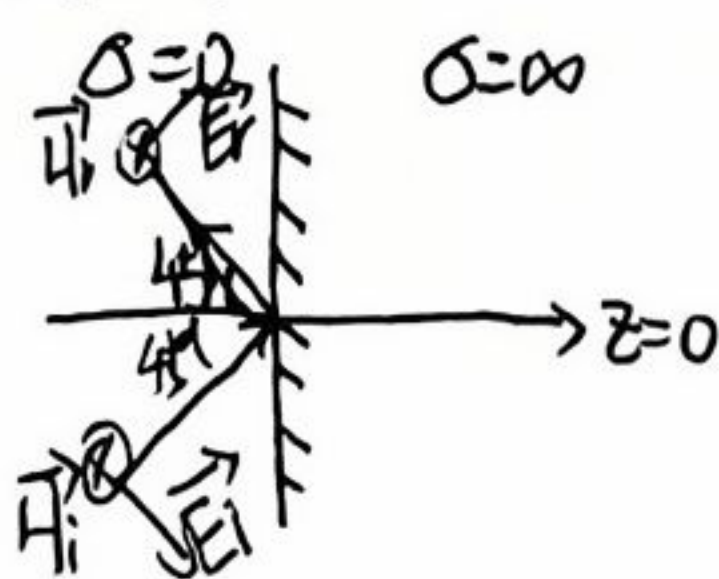
$$T_{\parallel} = \frac{E_{tm}}{E_{im}} = \frac{\eta_2 H_{tm}}{\eta_1 H_{im}} = \frac{\eta_2 (H_{im} + H_{rm})}{\eta_1 H_{im}} = \frac{\eta_2}{\eta_1} (1 + \Gamma_{\parallel})$$

$$= \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

$$\mu_1 = \mu_2 = \mu \quad \Gamma_{\parallel} = \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} \quad T_{\parallel} = \frac{2 \sin \theta_t \cos \theta_i}{\sin(\theta_i + \theta_t) \cos(\theta_i - \theta_t)}$$

例: 空气中平行极化波 $\vec{H}_i = -\vec{e}_y e^{-j\pi(x+z)}$ 分界面 $z=0$

求反射波电场



$$T_{\parallel} = 1$$

$$\vec{E}_r = \eta_0 \frac{\vec{S}}{2} (\vec{e}_x + \vec{e}_z) \times \vec{e}_i e^{j\vec{k}_r \cdot \vec{r}}$$

$$= \eta_0 \frac{\vec{S}}{2} (\vec{e}_x + \vec{e}_z) \cdot \vec{x} e^{-j\pi(x-z)}$$

$$\text{解: } \vec{k}_i = \sqrt{2}\pi \vec{e}_x + \sqrt{2}\pi \vec{e}_z$$

$$\vec{e}_i = \frac{\sqrt{2}}{2} (\vec{e}_x + \vec{e}_z)$$

$$\vec{E}_i = \eta_0 \vec{H}_i \times \vec{e}_i$$

$$\text{传播方向: } \vec{e}_r = \frac{\sqrt{2}}{2} (\vec{e}_x - \vec{e}_z)$$

$$\text{电场方向 } \frac{\vec{E}_r}{|\vec{E}_r|} = \frac{\sqrt{2}}{2} (\vec{e}_x + \vec{e}_z)$$

2. TE波, TM波 (传播方向: 沿z轴)

TEM波 (横电磁波) 电场与磁场均垂直于传播方向

TE波 (横电波) 电场与传播方向垂直, 传播方向上有磁场分量

TM波 (横磁波) 磁场与传播方向垂直, 传播方向上有电场分量

3. 均匀平面波 $v = \frac{w}{k} = \frac{1}{\sqrt{\mu\epsilon}}$ $k = \frac{2\pi}{\lambda} = \omega\sqrt{\mu\epsilon}$

$$\mu_0 = 4\pi \times 10^{-7} \quad \epsilon_0 = \frac{1}{36\pi} \times 10^{-9}$$

极化方式

$$\text{良导体 } \alpha = \beta \approx \sqrt{\frac{\omega\mu\sigma}{2}} \quad \delta = \frac{1}{\alpha} \quad \bar{\eta} = \sqrt{\frac{\omega\mu}{\sigma}} e^{i\frac{\pi}{4}}$$

波印廷矢量

$$\vec{S} = \vec{E} \times (\vec{H}_1 - \vec{H}_2)$$

$$\text{电磁场能量密度 } w = w_e + w_m = \frac{1}{2} \vec{E} \cdot \vec{D} + \frac{1}{2} \vec{B} \cdot \vec{H} = \frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2$$

4. 边界条件

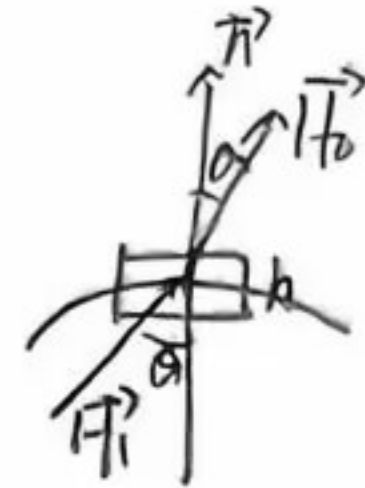
$$\vec{H}_{1t} - \vec{H}_{2t} = \vec{J}_s$$

$$E_{1t} - E_{2t} = 0$$

$$B_{1n} - B_{2n} = 0$$

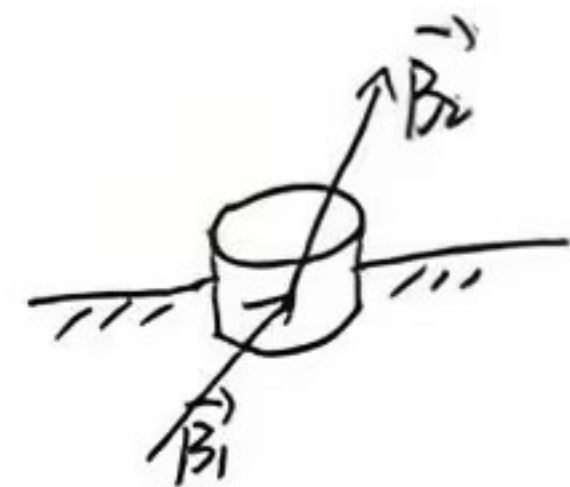
$$D_{1n} - D_{2n} = \rho$$

$$\begin{aligned} h \rightarrow 0 \quad \oint \vec{H} \cdot d\vec{l} &= \int_a^b \vec{H}_2 \cdot d\vec{l} + \int_b^a \vec{H}_1 \cdot d\vec{l} = J_s \Delta l \\ &= H_2 \sin \theta_2 \Delta l - H_1 \sin \theta_1 \Delta l \\ &= H_{2t} \Delta l - H_{1t} \Delta l = J_s \Delta l \end{aligned}$$



$$\begin{aligned} \oint \vec{E} \cdot d\vec{l} &= \int_1^2 \vec{E} \cdot d\vec{l} + \int_2^3 \vec{E} \cdot d\vec{l} + \int_3^4 \vec{E} \cdot d\vec{l} + \int_4^1 \vec{E} \cdot d\vec{l} \\ h \rightarrow 0 \quad &= \int_1^2 \vec{E}_1 \cdot d\vec{l} - \int_4^1 \vec{E}_2 \cdot d\vec{l} = 0 \end{aligned}$$

$$\begin{aligned} \oint \vec{B} \cdot d\vec{S} &= B_2 \cos \theta_2 \Delta S - B_1 \cos \theta_1 \Delta S \\ &= B_{2n} \Delta S - B_{1n} \Delta S = 0 \end{aligned}$$



$$\oint \vec{D} \cdot d\vec{S} = D_2 \cos \theta_2 \Delta S - D_1 \cos \theta_1 \Delta S = q = \rho \Delta S$$

5. 介质极化 $\vec{P} = \frac{\sum \vec{p}}{\Delta V}$ $\vec{P}$ 极化强度 $\vec{p}$ 电偶极矩

例: 圆柱形极化电介质极化强度沿电轴线方向, 高l, 半径a 均匀极化, 求束缚电荷及束缚面电荷分布

取柱坐标

$$\text{设 } \vec{P} = P_0 \vec{e}_z$$

$$\text{均匀极化 束缚体电荷 } \rho = -\nabla \cdot \vec{P} = 0$$



$$\begin{aligned} \text{侧面 } P_{sp} &= \vec{e}_r \cdot \vec{P} \\ &= \vec{e}_r \cdot P_0 \vec{e}_z \\ &= 0 \end{aligned}$$

$$\text{顶 } P_{sp} = \vec{e}_z \cdot \vec{e}_z P_0 = P_0$$

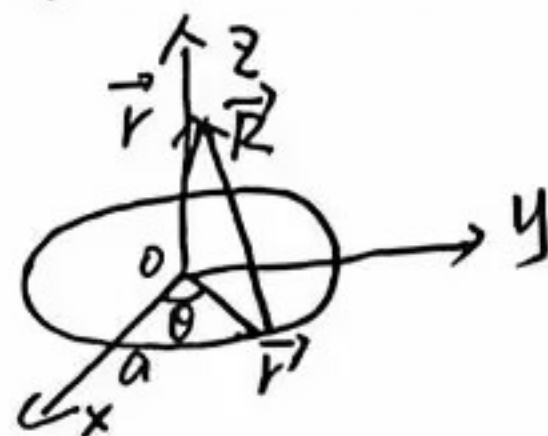
$$\text{底 } P_{sp} = -\vec{e}_z \cdot \vec{e}_z P_0 = -P_0$$

6. 已知 (体, 线, 面) 分布均匀的电荷, 求解静电场

例: 半径为a圆环, 求轴线上电场强度

$$\vec{E} = \frac{\int \rho dl}{4\pi\epsilon_0 r^2}$$

$$dl = a d\theta$$



$$\vec{r} = z \vec{e}_z$$

$$\vec{r}' = a \cos \theta \vec{e}_x + a \sin \theta \vec{e}_y$$

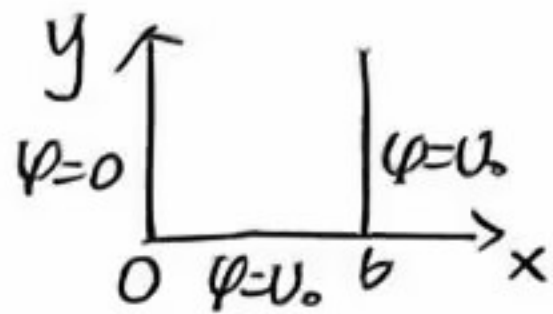
$$\vec{R} = \vec{r} - \vec{r}' = z \vec{e}_z - a \cos \theta \vec{e}_x - a \sin \theta \vec{e}_y$$

$$\vec{E}(\vec{r}) = \frac{\rho_l a}{4\pi\epsilon_0 (a^2 + z^2)^{3/2}} \int_0^{2\pi} (z \vec{e}_z - a \cos \theta \vec{e}_x - a \sin \theta \vec{e}_y) d\theta$$

7. 矢量磁位

$$\vec{B} = \nabla \times \vec{A} \quad \nabla^2 \vec{A} = -\mu \vec{J}$$

8. 分离变量法.



$$\varphi = \frac{V_0 x}{b} + \varphi_1$$

$$y \rightarrow 0 \quad \varphi_1 = -\frac{V_0 x}{b} + V_0$$

$$y \rightarrow \infty \quad \varphi_1 = 0$$

转化成求 φ_1 边界条件电位

$$\varphi = \frac{V_0 x}{b} + \varphi_1$$

本题电位与 z 无关, 只是 x, y 的函数, 即 $\varphi = \varphi(x, y)$

在 $0 < x < b, 0 < y < \infty$ 内 $\nabla^2 \varphi = 0$

边界条件为 ① $\varphi_1(0, y) = 0$

② $\varphi_1(b, y) = 0$

③ $\varphi_1(x, 0) = -\frac{V_0 x}{b} + V_0$

④ $\varphi_1(x, \infty) = 0$

$$\varphi_1(x, y) = X(x)Y(y)$$

由①②, $X(x)$ 为周期函数, 正弦形式 ($\lambda^2 < 0$)

$$X(x) = A_1 \sin k_x x + A_2 \cos k_x x$$

代入① $A_2 = 0$ 代入② $k_x = \frac{n\pi}{b}$

$$X(x) = A_1 \sin \frac{n\pi x}{b}$$

$$\lambda^2 + \beta^2 = 0 \quad \beta^2 > 0 \quad Y(y) = C_1 e^{-\frac{n\pi y}{b}} + C_2 e^{\frac{n\pi y}{b}}$$

$$\text{代入④} \quad C_2 = 0 \quad \varphi_n = \frac{A_1 C_1}{C_n} \sin \frac{n\pi x}{b} e^{-\frac{n\pi y}{b}}$$

$$\text{取不同的 } n \text{ 叠加} \quad \varphi_1(x, y) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi x}{b} e^{-\frac{n\pi y}{b}}$$

$$\text{代入③} \quad \varphi_1(x, 0) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi x}{b} = -\frac{V_0 x}{b} + V_0$$

$$\text{两边分别乘} \sin \frac{m\pi x}{b} \text{ 积分} \quad \frac{b}{2} C_n = \int_0^b (-\frac{V_0 x}{b} + V_0) \sin \frac{n\pi x}{b} dx$$

$$C_n = \frac{2V_0}{n\pi} \quad \varphi_1(x, y) = \sum_{n=1}^{\infty} \frac{2V_0}{n\pi} \sin \frac{n\pi x}{b} e^{-\frac{n\pi y}{b}} = \frac{V_0 b}{n\pi} \int_0^b x d \cos \frac{n\pi x}{b} - V_0 \frac{b}{n\pi} \cos \frac{n\pi x}{b} \Big|_0^b$$

$$\begin{aligned} \varphi(x, y) &= -\frac{V_0 x}{b} + \sum_{n=1}^{\infty} \frac{2V_0}{n\pi} \sin \frac{n\pi x}{b} e^{-\frac{n\pi y}{b}} \\ &= \frac{V_0}{n\pi} \left[x \cos \frac{n\pi x}{b} \Big|_0^b - \int_0^b \cos \frac{n\pi x}{b} dx - \frac{bV_0}{n\pi} (\cos n\pi - 1) \right] \\ &= \frac{V_0}{n\pi} (b \cos n\pi - b) - \frac{bV_0}{n\pi} \cos n\pi + \frac{bV_0}{n\pi} \\ &= \frac{bV_0}{n\pi} \end{aligned}$$

9. 基于麦克斯韦方程组求解电场、磁场、位移电流及平均波印廷矢量

大题: ① 斜入射

② 边界条件证明

③ 分离变量

④ 六七.

⑤ 场强

⑥ 麦.

小题: ① TE, TEM, TM.

② $\vec{P}$ $\vec{J}$

③ $\nabla \times \vec{A}$

④ 静电比拟